Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Features captured per purchase

37

Missing Values

Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

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Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

02

Initial Exploration

Checked structure with df.info() and summary statistics with .describe()

03

Missing Data Handling

Imputed 37 missing review ratings using median by product category

04

Column Standardization

Rename columns to snake case for better readability

05

Feature Engineering

Created age_group and purchase_frequency_days columns

06

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

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Revenue by Gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Key Finding

Revenue comparison between male and female customers revealed spending patterns that can inform gender-targeted marketing strategies.

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customer_id bigint	purchase_amount bigint
2	
3	
4	
7	
9	
12	
13	
16	
20	
22	

tal rows: 839 Query complete 0

High-Spending Discount Users

Identified customers who used discounts but still spent above average purchase amounts, revealing valuable insights into discount effectiveness and customer purchasing power.

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Top Products by Rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Product Excellence

Products with highest average review ratings identified for promotional campaigns.

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Shipping & Subscription Insights

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping options.

Subscriber Analysis

Compared average spend and total revenue across subscription status.

customers

10

28

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Customer Segmentation

New

First-time buyers

Returning

Multiple purchases

Loyal

Consistent shoppers

Classified customers based on purchase history to identify repeat buyers and subscription patterns.

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Power BI Dashboard

Interactive dashboard built in Power BI to present insights visually, enabling stakeholders to explore data and make informed decisions.

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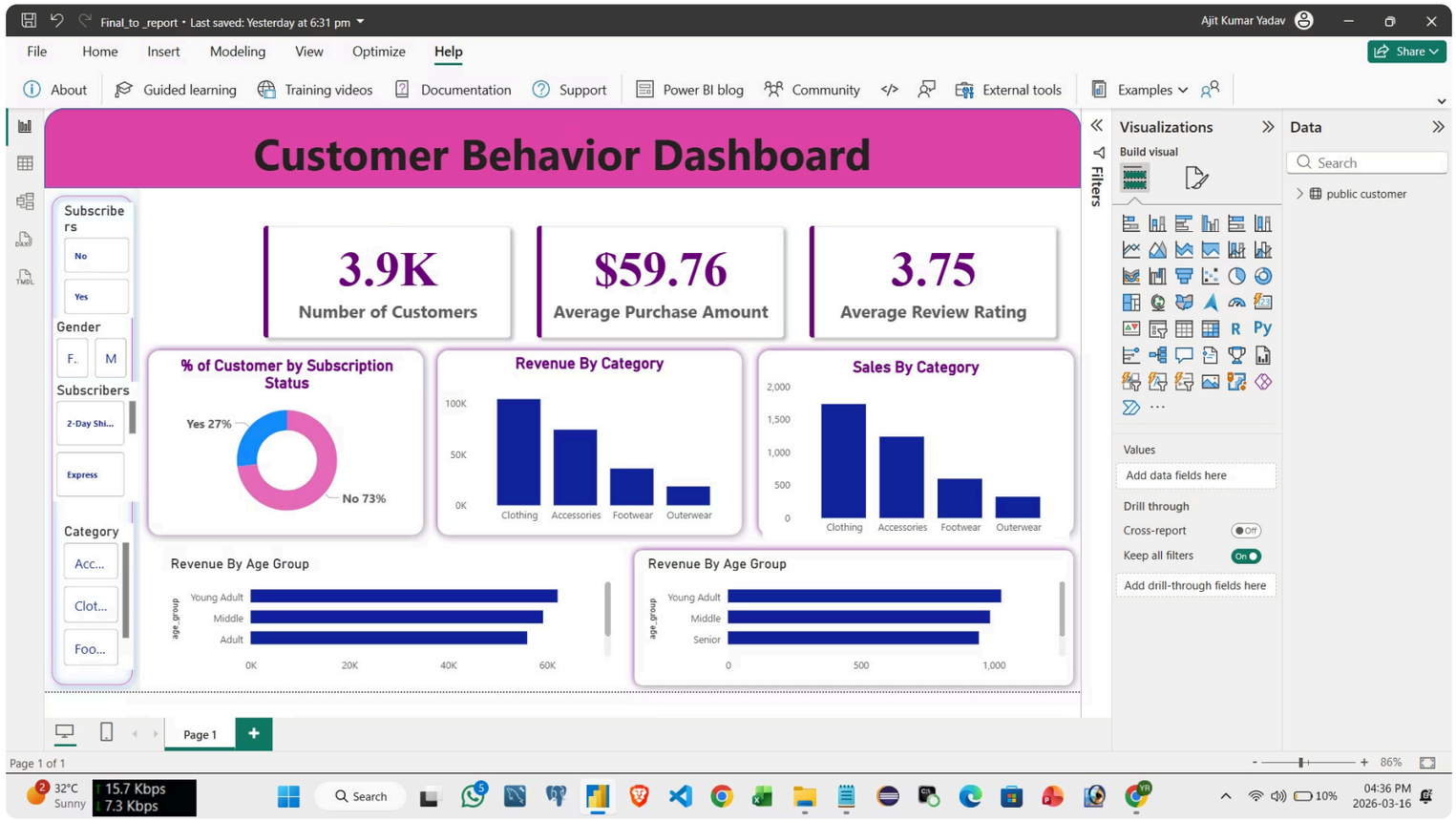

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to increase retention

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users

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